Project Summary

Harmonizing FreeMoCap Movement Data for Rehabilitation Research

Free and open-source motion capture can expand access to quantitative movement measurements for rehabilitation researchers and investigators working in clinical settings. However, wider access to data collection does not ensure that the resulting measurements can be understood and reused across studies. Differences in anatomical definitions, pose-estimation methods, coordinate conventions, and processing can affect interpretation. These distinctions must remain available when data leave the software that produced them.

This Data STACHE project will harmonize existing FreeMoCap gait and balance data with the DAPR framework and build that harmonization into a reusable FreeMoCap export. The work will connect movement time series with study context, measurement definitions, and processing history. Existing standards, including Motion-BIDS, will inform the design. Consultation with the DAPR Resource Core will guide terminology mappings and proposals for information not adequately represented in current resources.

Aim 1 will harmonize existing validation data in which unimpaired adults performed treadmill walking across speeds and the Modified Clinical Test of Sensory Interaction on Balance (CTSIB-M), preserving the context needed to interpret measurements from different pose-estimation backends and the marker-based reference. The resulting data packages, dictionary, and mappings will establish the requirements for Aim 2. Aim 2 will implement metadata capture and storage within the supported FreeMoCap workflow and an export capability that carries this information with the numerical data. Evaluation will check that exports reproduce the established representation, preserve measurements, and support representative reuse examples.

Within 12 months, the project will deliver harmonized gait and balance datasets, documented export software, reviewed mappings, and an evaluation report with worked examples. These outputs will support a methods manuscript and a follow-on study in rehabilitation cohorts. Because the same export is available to every FreeMoCap user, including rehabilitation and clinical groups, the pilot provides a direct route from new recordings to harmonized data.

Project Narrative

Affordable, open-source motion capture can make quantitative movement measurements accessible to more rehabilitation researchers, including investigators working in clinical settings. This project will harmonize existing FreeMoCap gait and balance data with shared terminology and the information needed to understand how measurements were obtained. Incorporating that harmonization into a reusable FreeMoCap export will help researchers prepare interpretable movement data for sharing and future use across studies.

Specific Aims

Harmonizing FreeMoCap Movement Data for Rehabilitation Research

Movement measurements are central to understanding walking and balance, yet their reuse across studies is limited by differences in data organization and measurement definitions. A variable called knee angle may reflect different landmarks, segment axes, or rotation sequences. Likewise, a trajectory may represent a surface marker, an estimated joint center, or a pose-estimation landmark. Pooling these measurements without their context can obscure physiological differences or mistake measurement differences for biological variation. Accessible systems such as FreeMoCap increase opportunities to collect movement data, making interoperable outputs and explicit provenance increasingly important.

FreeMoCap is a free and open-source markerless motion capture system that produces three-dimensional full-body kinematic data from consumer cameras [1], with over 10,000 GitHub stars and at least 15,000 users across 153 countries. Published applications span reaching and grasping biomechanics [2], gait analysis [3], and robotic exoskeleton design and verification [4]. Independent studies have evaluated FreeMoCap against marker-based motion capture for selected tasks and outcomes [5,6].

This breadth of use creates an opportunity to make harmonized outputs available to researchers at the point of export. Our validation work compared three pose-estimation backends against simultaneous marker-based measurements as unimpaired adults performed treadmill gait and the CTSIB-M balance test [7]. Agreement depended on the estimator, the task, and the outcome; backends that performed well for gait did not necessarily perform best for balance. If validity is conditional on how a measurement was produced, those conditions must travel with the data.

The objective of this Data STACHE pilot is to harmonize existing gait and balance data with the DAPR framework and incorporate that harmonization into a reusable FreeMoCap export. We will work with the Resource Core to map relevant information to existing terminology and identify metadata needs for movement time series. Existing resources, including Motion-BIDS, will inform the representation and organization of the data. Harmonization of the validation datasets will establish the requirements for software that other researchers can use with supported FreeMoCap recordings.

Aim 1 Harmonize existing FreeMoCap gait and balance validation datasets. We will inventory the study's movement measurements and the information needed to interpret them, including task context, measurement definitions, and acquisition and processing provenance, along with context fields that clinical recordings will require. With DAPR consultation, we will establish a data dictionary and terminology mappings, draw on applicable standards, and document unresolved metadata needs. We will assemble harmonized datasets and evaluate preservation of numerical values and measurement meaning. The outcome will be documented gait and balance data packages that establish the representation and metadata requirements for the export.

Aim 2 Build the harmonized representation into a reusable FreeMoCap export. We will implement the architecture needed to capture and retain relevant metadata during processing, connect researcher-supplied study context to recordings, and export numerical data with their definitions and provenance. Using the Aim 1 datasets as a reference, we will evaluate exports across the supported tasks and processing methods, check handling of missing metadata, and demonstrate representative reuse. The outcome will be tested export software and documentation that enable other researchers to prepare DAPR-aligned outputs from supported FreeMoCap recordings.

Within 12 months, the pilot will deliver harmonized gait and balance datasets, reviewed mappings, the export capability, and an evaluation report. Because the export is part of free, open-source software, any lab using FreeMoCap, including rehabilitation and clinical groups, can produce DAPR-aligned data from its own recordings. Findings will inform DAPR's handling of movement time series and lay the groundwork for a methods manuscript and a follow-on study in rehabilitation cohorts.

Research Strategy

Significance and Preliminary Work

Biomechanical measurements can vary depending on the models and assumptions used to derive them. For example, in gait, anthropometric model selection can alter estimated mediolateral and vertical center-of-mass motion [8]. In a 2025 ISB community consultation, 46% of respondents identified insufficient methodological detail as the biggest obstacle to comparing joint-motion data across research groups [9]. Preserving these details alongside numerical outputs would help researchers assess which measurements are suitable for reuse and comparison.

FreeMoCap makes quantitative movement measurements accessible through consumer cameras and free, open-source software [1]. Its use across research settings creates an opportunity to improve how movement data are documented and shared. For rehabilitation research, reuse requires information about the protocol, measurement definitions, and processing methods. The same quantity can be stored in non-equivalent forms, and systems differ in how much processing precedes output [10]; a familiar variable name does not establish comparability.

Our gait and balance validation compares FreeMoCap outputs from MediaPipe, RTMPose, and ViTPose with Qualisys marker-based measurements [7]. Figure 1 shows estimator-dependent differences in sagittal gait kinematics, and across the full study, the backend that performed best for gait was not necessarily best for standing balance. Developing this validation workflow has also required us to specify anatomical definitions and joint-angle conventions that are not conveyed by the resulting numerical arrays. This experience provides a practical basis for identifying the metadata that should accompany shared movement data and for evaluating whether harmonized exports preserve that information.

DAPR's development priorities include infrastructure for gait time series [11]. Motion-BIDS organizes raw motion time series and describes their reference frames, but it leaves derived data such as joint angles unstandardized and does not prescribe anatomical nomenclature or landmark definitions [10]. Those details determine how joint angles and center-of-mass estimates should be interpreted, and they are what this pilot adds. Harmonizing the existing data will establish metadata requirements; incorporating them into FreeMoCap will help other researchers retain that information during processing and export.

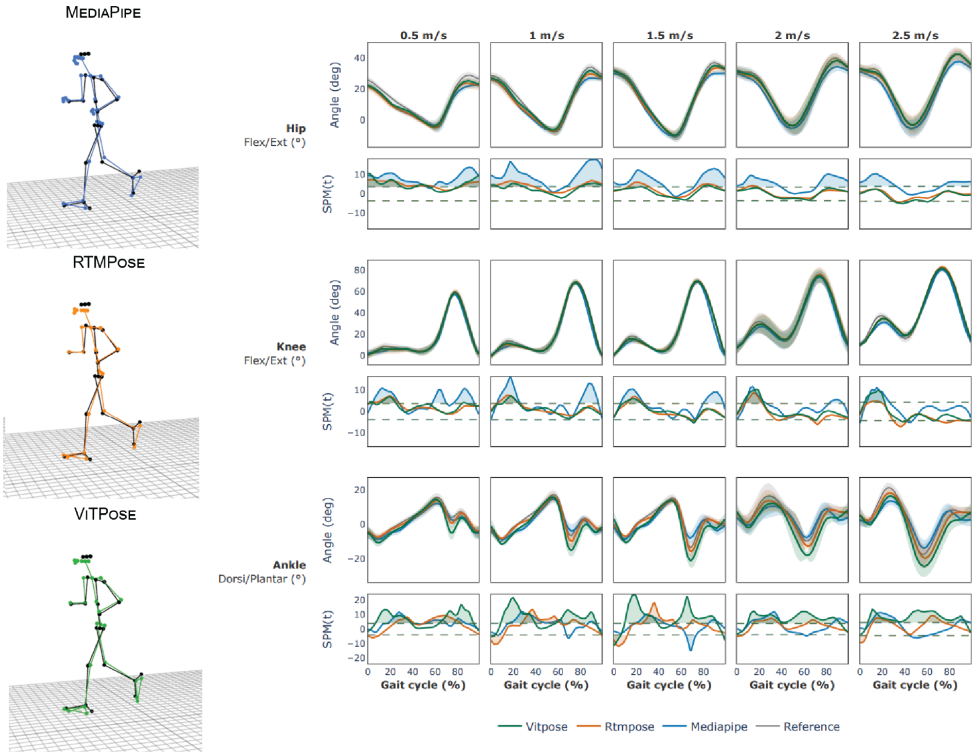


Figure 1: **Existing FreeMoCap gait validation.** Left: example reconstructions from three pose-estimation backends. Right: sagittal hip, knee, and ankle kinematics across five treadmill speeds (0.5-2.5 m/s), with Qualisys as the gray reference. Lower rows show statistical parametric mapping (SPM) t-statistics for comparisons with the reference. These comparisons motivate preserving tracking methods, measurement definitions, and processing history.

Approach

Study Design and Existing Data

The project will use the existing FreeMoCap validation release, comprising six participants and 24 treadmill gait and standing balance trials, with aligned trajectories from MediaPipe, RTMPose, ViTPose, and Qualisys [12]. Participants were unimpaired adults, a deliberate choice for a first harmonization effort. Treadmill walking at controlled speeds and the CTSIB-M are standard in rehabilitation assessment, and the slowest speeds overlap those common in clinical populations. Because each trial was measured simultaneously by three pose-estimation backends and a marker-based reference, differences between measurement methods can be separated from differences between people; in pooled clinical data, the two are confounded unless the method is recorded with the data. The dictionary will also include rehabilitation context fields, such as assistive device, orthosis or prosthesis, handrail or harness use, and affected side, recorded as not applicable for this cohort so the export can carry them for clinical recordings.

The validation study provides both shared movements observed through different measurement methods and distinct gait and balance task contexts. This combination allows us to evaluate whether a common representation preserves differences among methods while describing task-specific information consistently. Existing analysis code and configuration records will provide reference calculations and help identify metadata that FreeMoCap should retain systematically for future recordings.

Aim 1: Harmonize Existing Gait and Balance Validation Datasets

In months 1-3, we will inventory the selected data and source records, document versions and permissions, and define representative reuse analyses. The dictionary will cover study and coded participant identifiers, sessions and trials, task conditions, measurement definitions, units, timing, coordinate frames, and acquisition and processing provenance. We will distinguish information available from the software workflow from context supplied through protocol documentation, and specify required and optional fields for each supported data type. Missing information will remain explicit.

Metadata requirements will reflect the measurements being shared. Joint-angle records will describe segment definitions, anatomical axes, sign conventions, and the rotation sequence or projection method, referencing applicable ISB recommendations and documenting the conventions actually used [13]. For center-of-mass estimates, we will identify the anthropometric model and implemented parameters, including segment definitions, mass fractions, and relative segment COM locations [14]. Other records will identify the pose estimator, model and software versions, filtering, interpolation, and analysis intervals. These examples will guide a common metadata core with additional fields for particular measurements. Definitions and parameters will be machine-readable and linked to the relevant numerical outputs.

With the DAPR Resource Core, we will review mappings to DAPR and established terminology, including NINDS or OMOP concepts where their definitions apply. Each mapping will retain the source definition, target identifier and version, and any transformation. Exact matches, matches requiring transformation, related concepts, and unresolved mappings will remain distinguishable. The PI and mentor will review mappings used in the demonstrations; missing concepts will be proposed to DAPR with definitions and examples. Motion-BIDS will inform applicable conventions for signals, channels, reference frames, and data organization, with versions, adopted conventions, and additional metadata documented explicitly.

Consultation with the Resource Core will begin by confirming the initial scope and receiving requirements. Intermediate review will address the dictionary, mapping decisions, and representative exports; final review will address the completed packages, remaining limitations, and handoff or import procedures.

In months 3-5, we will assemble harmonized gait and balance packages that link numerical data, definitions, provenance, and source records through stable identifiers. Files will use documented, openly readable formats; Numerical tables will use Parquet, the FreeMoCap 2.0 output format, with linked metadata files. Original values and missingness will be preserved, and any unit or coordinate transformation will be explicit. Original timestamps, synchronized data, and waveforms normalized to a gait cycle will remain distinguishable.

Evaluation will account for each selected source field as represented or excluded with a reason. Coverage will be reported by task, data type, and measurement method, distinguishing available information, recovered information, and unresolved gaps. Representative analyses will test whether the attached definitions and inputs support reuse. By month 5, the packages, dictionary, mappings, and initial coverage and fidelity report will establish the reference representation for Aim 2.

Research Strategy Continued

Aim 2: Build the Harmonized Representation into a Reusable FreeMoCap Export

In months 5-9, we will incorporate the Aim 1 requirements into FreeMoCap's processing and export workflow. The exporter targets FreeMoCap 2.0, whose pipeline separates raw, processed-data, and export zones, writes Parquet outputs, and saves intermediate 2D detections and raw 3D reconstructions so each stage can be restarted independently. A restartable stage must already record its inputs and parameters, so provenance capture extends the existing design rather than adding a parallel system. Recordings and derived measurements will be associated with versioned metadata records that retain relevant information through processing stages. Available software versions, pose-estimation settings, and processing parameters will be captured from supported workflows; measurement definitions and calculation settings will be linked to their outputs. Information recovered retrospectively in Aim 1 will identify what future recordings need to store systematically.

Researchers will supply context that software cannot infer, such as task conditions and protocol descriptions, through a documented recording or study manifest. The exporter will combine this context with stored processing metadata and produce harmonized numerical and metadata files. Checks will identify missing required fields, inconsistent descriptions, and unsupported data types. An export report will describe supported uses and unresolved gaps.

In months 8-10, we will compare exports with the Aim 1 reference packages across supported gait and balance tasks and processing methods. Automated checks will assess numerical fidelity, identifiers, units, timing, measurement definitions, and provenance. Constructed test cases will evaluate metadata capture and propagation, researcher-supplied context, and handling of missing essential fields or incompatible definitions. Discrepancies will be investigated before release. Evaluation will distinguish file validity from sufficient documentation for a particular analysis.

Worked examples will reproduce selected analyses from exported data using documented environments and instructions. The PI will run them in a clean environment and compare results with native analyses using predefined tolerances. The mentor will review representative exports, instructions, and results for clarity and consistency with measurement definitions. Findings will guide revisions to the exporter and documentation, with release materials completed in months 10-12.

Rigor, Feasibility, and Responsibilities

Results will be descriptive and reported by task and measurement method, retaining shared participant and trial identities. Versioned source files, mappings, transformation parameters, and software environments will support reproducibility. Aaron T. Cherian will lead harmonization, metadata architecture, implementation, evaluation, and DAPR coordination. Jonathan Samir Matthis, PhD, President and CEO of the FreeMoCap Foundation, will provide mentorship and review of scientific assumptions, mappings, software design, and findings. Monthly meetings and checkpoints at months 3, 6, 9, and 12 will track progress.

The initial release will support defined FreeMoCap versions, data types, and metadata requirements established from the validation datasets. The exporter will be merged into FreeMoCap through standard code review and released as a standalone archived package. Optional metadata and graphical interface enhancements may be deferred to protect core functionality.

Limitations and Alternatives

Evaluation within one validation study will establish performance for its supported data; transfer to independent cohorts and sites will require later assessment. Data come from unimpaired adults; estimator performance and required context in clinical populations will be assessed in follow-on work. Missing information and meaningful differences among pose estimators and anatomical models will remain visible. DAPR consultation will guide terminology and receiving requirements, with local definitions retained for unresolved terms. If central ingestion is unavailable, we will provide versioned packages, mappings, and a documented handoff. Harmonized representation does not establish clinical validity or make different measurements interchangeable.

Dissemination, Sharing, and Future Research Plan

The project will release harmonized gait and balance datasets, a data dictionary and terminology mappings, reusable FreeMoCap export software, and an evaluation report. Documentation will explain the supported inputs, metadata captured and stored during processing, study context supplied by the researcher, and known limitations. Worked examples will show how to prepare an export, inspect measurement definitions, select relevant records, and reproduce representative analyses.

By month 10, we will prepare a release candidate for DAPR review. By the end of the award, permitted data, metadata, code, and evaluation materials will be archived in an established repository such as Zenodo, with persistent identifiers and source attribution. GitHub will support development and issue reporting. Source licenses will be preserved, and software incorporated into FreeMoCap will follow its applicable licensing requirements. Scientific data will be shared by publication or, for other findings, by the award's end, consistent with the Data Management and Sharing Plan.

A short tutorial and an offered DAPR presentation will demonstrate the export and its use with the harmonized datasets. The exporter will ship in a FreeMoCap release and as a standalone archived package that runs independently of any particular application release. Feedback received through demonstrations and repository issues will inform future improvements. Archived releases and versioned documentation will preserve the pilot's results as the software evolves.

We plan to submit at least one peer-reviewed methods or research-software manuscript describing the representation and its evaluation. The paper will address the metadata needed to interpret gait and balance time series, how existing standards and DAPR terms cover those needs, and how metadata capture and export preserve that information for reuse. We will prepare the manuscript outline in months 10–11 and submit after the award ends.

In the second half of the award, we will consult NCMRR program staff, identify clinical-research collaborators through DAPR and the other Medical Rehabilitation Research Resource Centers, and draft follow-on specific aims. We anticipate an NIH R21, or a mechanism program staff recommend, applying the export to rehabilitation cohorts and testing whether estimator performance and metadata requirements differ in clinical populations. The pilot's coverage results and documented metadata gaps will define its questions.

The PI will coordinate the six-month progress report, final scientific and expenditure reports, and required annual follow-up reporting. The final report will identify the harmonized datasets, supported exports, evaluation results, and remaining work needed for broader adoption.

Project Timeline

The project runs for 12 months from the approved start date. Work will progress from harmonization of the existing gait and balance datasets to implementation and evaluation of the reusable FreeMoCap export.

Period	Activities	Milestone and deliverable
Months 1-2	Complete institutional and data-use review; freeze source release; inventory gait and balance outputs and associated metadata.	Source manifest; documented permissions; initial scope of supported data and reuse examples.
Month 3	Review the dictionary, existing standards, DAPR mappings, and metadata available from processing or protocol documentation.	Representation version 0.1; mapping decisions; required and optional fields; unresolved terms logged.
Months 4-5	Assemble harmonized gait and balance packages; assess coverage, numerical fidelity, and representative reuse.	Aim 1 datasets, dictionary, mappings, and initial evaluation; requirements for metadata storage and export.
Month 6	Implement recording and measurement metadata records and their links through processing; review progress.	Initial metadata architecture and capture mechanisms; six-month progress report and budget update.
Months 7-8	Implement study-context input, export generation, and checks; compare initial exports with Aim 1 packages.	Runnable export; supported-input documentation; initial fidelity and metadata propagation results.
Month 9	Test supported tasks and methods, missing metadata, and clean-environment reuse; conduct mentor review and assess DAPR handoff needs.	Reuse checks and mentor review log; revised software and documentation; follow-on concept outline.
Months 10-11	Prepare release candidate, tutorial, evaluation report, and manuscript outline; discuss future funding.	DAPR review materials; organized results; draft follow-on aims and software contribution for review.
Month 12	Archive permitted outputs; demonstrate the export; complete reporting.	Persistent identifiers for datasets, exporter, dictionary, mappings, tutorial, and final reports.
Year after completion	Planned manuscript submission and application to an appropriate follow-on opportunity.	Target: Methods manuscript submission and a follow-on application (R21 or mechanism recommended by program staff)

The PI will carry out harmonization, software development, and evaluation, with monthly mentor review and quarterly checkpoints.

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Resources

Institutional Environment and Infrastructure

The FreeMoCap Foundation is a nonprofit organization responsible for the development and stewardship of FreeMoCap, an open-source platform for accessible markerless motion capture. The proposed work will make a harmonized export capability available for supported FreeMoCap outputs. Familiarity with the software and its development process provides a practical basis for implementing the workflow and documenting how researchers can use it.

Aaron T. Cherian, PhD, is the proposed PI and will carry out data harmonization, metadata architecture, export implementation, checks, and documentation. His experience with markerless motion capture, gait and balance analysis, and comparisons among tracking methods will inform metadata selection and measurement definitions. He will coordinate DAPR consultation, lead evaluation, and develop dissemination and follow-on research plans.

Jonathan Samir Matthis, PhD, President and CEO of the FreeMoCap Foundation, will serve as project mentor. He will review measurement definitions and mappings, advise on metadata architecture and export design, and review evaluation results and release documentation. The budget supports the PI and mentor. The PI will run reproducibility checks in a clean environment; mentor review will identify questions or gaps in the examples and documentation. The initial release will support defined FreeMoCap versions and data types, the existing gait and balance datasets, and representative analyses.

Development will use existing computers and the team's repository workflow, with modest funds for storage . Existing reconstructed data provide the necessary inputs. The project does not require new recording equipment or recruitment.

Data and Computational Resources

The data source is the public FreeMoCap validation release and analysis repository. It provides aligned trajectories from multiple pose-estimation backends and a marker-based reference, together with gait and balance analysis code. The data and existing analyses will support harmonization in Aim 1 and provide reference packages for evaluating the exporter in Aim 2. Measurements from the same trial will retain their relationship and method identities. The public release excludes raw videos and direct participant identifiers.

The gait and balance datasets belong to the same validation study and provide differing task contexts and measurement types for the pilot. Associated configurations, analysis code, and protocol documentation will help recover available metadata and identify information that FreeMoCap should retain systematically. The final source inventory will document the release, eligible records, and permissions for use and sharing. Broader application across independent studies will be enabled by the reusable export and assessed in subsequent work.

The implementation will add persistent metadata records and links through the supported FreeMoCap processing workflow, using documented, openly readable numerical and metadata formats with example loading and analysis code. Numerical tables will use Parquet. Existing analysis code will provide reference calculations where supported. The package definitions will be independent of any one internal software library so that other researchers can inspect and use the exported data.

Source files, working copies, metadata, and outputs will be connected through identifiers and checksums. Computation will focus on conversion, automated checks, and representative analyses using existing hardware. Code, configurations, dependencies, and relevant source versions will be retained together. Public releases will be archived in an established repository.

Resources Continued

DAPR Resources and Existing Standards

We request Resource Core consultation to review the representation, map the selected study descriptors and measurements, and identify information that current terminology does not adequately capture. Initial consultation will address scope and receiving requirements. Intermediate review will examine the dictionary, mapping decisions, and representative exports. Final review will address study packages, limitations, and handoff or import procedures. Meeting dates and effort will be agreed with the Core.

DAPR describes a Schema, Dictionary, Thesaurus, and MAPR mapping application. Where appropriate, mapping tools may suggest candidate terms; the team will review definitions, units, and relationships before accepting a match. Broader or related concepts will remain distinguishable from exact mappings. Terms without an adequate match will retain clear definitions and be proposed for DAPR review.

Existing resources, including Motion-BIDS, will inform how motion signals, channels, reference frames, and related metadata are represented. Applicable conventions will be adopted and the scope of conformance documented. Additional metadata will remain explicit, with source links and a rationale for its inclusion. This will allow the export design to draw on established work while responding to the needs identified in the study inventory and DAPR consultation.

The contribution to DAPR will include harmonized gait and balance datasets, reviewed mappings, a reusable export workflow, and an evaluation of the information needed for interpretation and reuse. If a central receiving interface is unavailable during the award, we will provide versioned packages and a documented handoff.

Versioned releases and accompanying documentation will be archived in an established repository to preserve the pilot’s outputs and support their continued use. Future maintenance and expansion will follow the project’s established development processes.

Resource Stewardship and Career Development

The PI and mentor will use monthly reviews and quarterly checkpoints to assess progress against the project aims and timeline and address implementation or evaluation challenges. Matthis will provide scientific and technical guidance and support the PI’s development of manuscript materials and follow-on research aims. Collaboration with DAPR will develop the PI’s expertise in rehabilitation terminology, data standards, and harmonization across studies.

Data Management and Sharing Plan

1. Will there be maximum appropriate sharing of scientific data underlying peer-reviewed publications and other findings resulting from the work supported by this award (including preprints, refereed papers reported at conferences, and other findings)?

Yes.

2. Will the scientific data underlying peer-reviewed publications be shared by the time of publication, or for other findings, by the end of the period of performance, which includes no-cost extensions?

Yes.

3. Will shared scientific data be made available for at least as long as required by applicable data repository policies and/or journal policies?

Yes.

4. If you answered “no” to elements 1, 2, or 3, or if you anticipate that sharing will be limited in some other way, please describe these limitations and the ethical, legal, or technical factors for them. Your response should specify a particular reason(s) for limiting sharing.

No additional limitations to sharing the scientific data and associated metadata generated by this project are anticipated.

5. If scientific data derived from human research participants will be shared, will privacy, rights, and confidentiality of participants be protected as outlined in NOT-OD-22-213?

Yes.

Public releases will contain only permitted, nonidentifying numerical data and metadata.

6. In the table below, please list:

Expected Data Type	Established Repository or Example
Reconstructed movement trajectories, kinematic time series and summaries, measurement definitions, processing history, and permitted study metadata.	Zenodo, linked to the source validation study; harmonized gait and balance packages and mappings provided to DAPR.
Data dictionary and mappings, evaluation results, metadata capture and export software, example loading and analysis code, and reproducible environments.	Zenodo archival releases; GitHub for maintained source code and documentation.

7. For studies subject to the NIH Genomic Data Sharing Policy (GDS) (e.g., using NIH funds to generate large-scale human genomic data):

a. Will you share all large-scale human genomic and associated data in a NIH-designated repository according to the accelerated timelines expected in the GDS Policy?

Not Applicable. No genomic data are proposed.

b. Do you anticipate that when sharing you will be able to meet the expectations of the Institutional Certification in the GDS Policy?

Not Applicable. No genomic data are proposed.

Human Subjects Determination

Project title: Harmonizing FreeMoCap Movement Data for Rehabilitation Research
Principal Investigator: Aaron T. Cherian
Applicant Organization: FreeMoCap Foundation

Proposed Activities and Data Source

The project will use the publicly released FreeMoCap gait and balance validation dataset to develop and evaluate harmonized movement-data packages and a reusable FreeMoCap exporter. The participant-level inputs will consist of publicly available numerical trajectories, derived measurements, task descriptors, and associated study metadata. Software and methodological documentation will support the description of processing steps and measurement definitions.

The original dataset was collected under Northeastern University IRB approval. Northeastern University will not participate in the proposed project. No new participants will be recruited or contacted, no new recordings will be collected, and no intervention or clinical trial is proposed.

Proposed Exemption

For application purposes, the PI proposes that the project qualifies for Exemption 4 under 45 CFR 46.104(d)(4)(i), concerning secondary research using publicly available information. The data were collected for an earlier validation study and are now publicly available for reuse. The proposed work will reorganize and document those data, evaluate the preservation of measurements and metadata during export, and develop reusable software. No additional nonpublic participant-level information will be obtained or incorporated into the project.

This exemption designation represents the PI's preliminary assessment for the application. A formal determination for the proposed Foundation-based project has not yet been obtained.

Identifiability and Privacy Protections

The public dataset omits direct participant identifiers. The PI retains access to information that can connect participant codes to identities; accordingly, this justification does not rely on an assertion that the PI cannot identify participants.

Participant linkage keys, names, contact information, consent records, and identifiable source videos will not be included in the project's analytical workspace or public outputs. The project will not link the research data to participant identities or other private records. Data use and sharing will follow the permissions applicable to the source dataset.

Confirmation Before Project Initiation

The Foundation will coordinate with DAPR to identify an appropriate independent authority to confirm the proposed classification and obtain any required determination or approval before beginning the funded secondary data analyses. The human subjects documentation will be updated to reflect that determination. Any proposed expansion to nonpublic participant data or new participant interaction will be assessed before implementation.

Program Director/Principal Investigator (Last, First, Middle): Cherian, Aaron, T.

DETAILED BUDGET FOR INITIAL BUDGET PERIOD
DIRECT COSTS ONLY

FROM: TBD at award | THROUGH: End of month 12

List PERSONNEL *(Applicant organization only)*
Use Cal, Acad, or Summer to Enter Months Devoted to Project
Enter Dollar Amounts Requested *(omit cents)* for Salary Requested and Fringe Benefits

NAME	ROLE ON PROJECT	Cal. Mnths	Acad. Mnths	Summer Mnths	INST.BASE SALARY	SALARY REQUESTED	FRINGE BENEFITS	TOTAL
Aaron T. Cherian	PD/PI	3.51			60,000	17,538	3,508	21,046
Jonathan S. Matthis	Mentor	1.17			80,000	7,795	1,559	9,354
				–				
SUBTOTALS →						25,333	5,067	30,400

CONSULTANT COSTS	0
EQUIPMENT *(Itemize)*	0
SUPPLIES *(Itemize by category)*	0
TRAVEL	0
INPATIENT CARE COSTS	0
OUTPATIENT CARE COSTS	0
ALTERATIONS AND RENOVATIONS *(Itemize by category)*	0
OTHER EXPENSES *(Itemize by category)* Project storage and backup: $50/month x 12 months	600
CONSORTIUM/CONTRACTUAL COSTS — DIRECT COSTS	0
SUBTOTAL DIRECT COSTS FOR INITIAL BUDGET PERIOD *(Item 7a, Face Page)*	$ 31,000
CONSORTIUM/CONTRACTUAL COSTS — FACILITIES AND ADMINISTRATIVE COSTS	0
TOTAL DIRECT COSTS FOR INITIAL BUDGET PERIOD	$ 31,000

Budget Justification

Project period: 12 months from the approved start date.

Requested direct costs: $31,000. Requested indirect costs: $2,480. Total request: $33,480.

The budget supports harmonization of existing gait and balance data, metadata capture and export within FreeMoCap, and evaluation. No funds are requested for recruitment, new recording equipment, model training, travel, or clinical services. The accompanying PHS 398 Form Page 4 presents the direct costs.

Personnel

Principal Investigator Aaron T. Cherian. Requested support is 3.51 calendar months, equivalent to 29.23% effort over 12 months. At a proposed salary base of $60,000, salary requested is $17,538. Provisional fringe at 20% adds $3,508, for a total of $21,046. The PI will carry out harmonization, metadata architecture, export implementation, automated checks, reuse demonstrations, and documentation. He will lead DAPR coordination, evaluation, manuscript preparation, and follow-on research planning.

Mentor Jonathan Samir Matthis, PhD. Matthis is President and CEO of the FreeMoCap Foundation. Requested support is 1.17 calendar months, equivalent to 9.74% effort over 12 months. At a proposed annual salary base of $80,000, salary requested is $7,795. Provisional fringe at 20% adds $1,559, for a total of $9,354. He will provide scientific and technical mentorship, review definitions and mappings, advise on architecture and evaluation, and review representative exports and documentation. Monthly meetings and quarterly checkpoints will also support manuscript and follow-on proposal development.

These two personnel lines total $30,400: $25,333 in salaries and $5,067 in provisional fringe. The PI's development effort and Matthis's mentorship will span harmonization, implementation, evaluation, and release.

Fringe Benefits and Compensation Assumptions

The proposed 20% fringe rate comprises employer Social Security and Medicare contributions at 7.65%, an estimated health-insurance contribution of 9.35%, and an estimated retirement contribution of 3.00%, applied to requested salaries.

The personnel budget assumes employee compensation during the award, with fringe benefits estimated at 20% of requested salary.

Other Direct Costs

Project storage and backup $600. Twelve months at a proposed average of $50 per month will support working copies, backup, and retained release candidates for the selected numerical data and derived products. These costs support source preservation, reproducibility, and data management. Final capacity and procurement will be matched to the frozen source inventory; public repository deposition will use available no-cost options where appropriate.

Indirect Costs and Budget Control

Indirect costs are calculated at 8% of $31,000 in direct costs, yielding $2,480 and a total request of $33,480. Project expenditures will be reviewed at the six-month reporting checkpoint and reconciled at close-out.

FreeMoCap Foundation, Inc.
A 501(c)(3) nonprofit organization

15 October 2026

DAPR Pilot Grant Program
University of Southern California

Re: Harmonizing FreeMoCap Movement Data for Rehabilitation Research

Dear DAPR Pilot Grant Review Committee:

On behalf of the FreeMoCap Foundation, I confirm the Foundation's support for this application and its willingness to enter into a formal subaward or consortium agreement with the University of Southern California if the project is selected for funding.

Principal Investigator: Aaron T. Cherian
Collaborating institutions: FreeMoCap Foundation and University of Southern California
Period of support: 12 months from the approved project start date

Direct costs	Indirect costs	Total request
$31,000	$2,480	$33,480

The proposed project will harmonize existing FreeMoCap gait and balance validation datasets and incorporate metadata capture and harmonized exports into FreeMoCap. The Foundation will provide administrative oversight for the approved scope of work, manage project funds, and meet applicable reporting, research, data-management, and sharing requirements.

If selected, the Foundation will work with USC to finalize the scientific, administrative, and financial arrangements and complete the formal agreement and any required assurances or certifications before award.

Sincerely,

Jonathan Matthis

Jonathan Samir Matthis
President and CEO, FreeMoCap Foundation
jon@freemocap.org

OMB No. 3145-0279 (Expiration Date 10/31/2026)

NIH BIOGRAPHICAL SKETCH COMMON FORM

Name: Cherian, Aaron T

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0003-4944-2814

Position Title: Chief Scientific Officer

Organization and Location: FreeMoCap Foundation, Boston, Massachusetts, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
Northeastern University, Boston, Massachusetts, United States	Doctor of Philosophy (PHD)	09/2021	05/2026	Bioengineering
Brown University, Providence, Rhode Island, United States	Master of Science (MS)	09/2018	05/2020	Biomedical Engineering
University of Maryland College Park, College Park, Maryland, United States	Bachelor of Science (BS)	09/2014	05/2018	Bioengineering

Appointments and Positions

2026 - present Chief Scientific Officer, FreeMoCap Foundation, Boston, Massachusetts, United States

2021 - 2026 Graduate Research Assistant, Northeastern University, Bioengineering Department, Boston, Massachusetts, United States

Products

Products Closely Related to the Proposed Project

1. Cherian AT. Open-Source Development and Validation of a Low-Cost Markerless System for Quantitative Motion Analysis. Boston, Massachusetts: Northeastern University; 2026 May. Available from: https://hdl.handle.net/2047/D20859645
2. Jonathan Samir Matthis, Philip Queen, Aaron Cherian, Trent Wirth, Endurance Idehen. FreeMoCap: A free, open source markerless motion capture system. 2026 August. Available from: https://zenodo.org/doi/10.5281/zenodo.22131400 DOI: 10.5281/ZENODO.22131400
3. Aaron Cherian. FreeMoCap Validation Dataset: Multi-Camera Markerless Motion Capture of Treadmill Gait and Standing Balance. Zenodo; 2026 September. Available from: https://zenodo.org/doi/10.5281/zenodo.23041223 DOI: 10.5281/ZENODO.23041223
4. Aaron Cherian, jonmatthis. SkellyForge: The 3D data processing backend for the FreeMoCap Project. 2023 December. Available from: https://zenodo.org/doi/10.5281/zenodo.10395649 DOI: 10.5281/ZENODO.10395649
5. jonmatthis, Endurance Idehen, philipqueen, Mac Prible, Aaron Cherian. SkellyCam: The camera backend for the FreeMoCap software. 2026 April. Available from: https://zenodo.org/doi/10.5281/zenodo.19921107 DOI: 10.5281/ZENODO.19921107

Other Significant Products Highlighting Contributions to Science

1. Nie J, Eom K, AlGhosain H, Neifert A, Cherian A, Gerbaka G, Ma K, Liu T, Lee J. Intravitreally Injected Plasmonic Nanorods Activate Bipolar Cells with Patterned Near-Infrared Laser Projection. ACS Nano. 2025 March 20; 19(12):11823-11840. Available from: https://pubs.acs.org/doi/10.1021/acsnano.4c14061 DOI: 10.1021/acsnano.4c14061; NIHMSID: NIHMS2081004
2. philipqueen, jonmatthis. SkellyTracker: The image analysis backend for the FreeMoCap Project. 2023 December. Available from: https://zenodo.org/doi/10.5281/zenodo.10395641 DOI: 10.5281/ZENODO.10395641
3. Aaron Thomas Cherian. Development and Optimization of an Intrinsic Optical Imaging System for Hemodynamic Correction of Fluorescent Signals. 2020 May. Available from: https://repository.library.brown.edu/studio/item/bdr:1129219/ DOI: 10.26300/XHJ5-KV66
4. philipqueen, jonmatthis, Aaron Cherian. SkellyClicker: For labeling and traing multi-camera video datasets. 2026 April. Available from: https://zenodo.org/doi/10.5281/zenodo.19389889 DOI: 10.5281/ZENODO.19389889

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§287, 1001, 1031 and 31 U.S.C. §§3729-3733 and 3802.

Certified by Cherian, Aaron T in SciENcv on 2026-10-16 13:06:21

NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Cherian, Aaron T

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0003-4944-2814

Position Title: Chief Scientific Officer

Organization and Location: FreeMoCap Foundation, Boston, Massachusetts, United States

Personal Statement

I am a researcher and Chief Scientific Officer of the FreeMoCap Foundation, where my work focuses on the development, validation, and application of accessible, open-source motion capture technologies. I received my PhD in Bioengineering from Northeastern University in 2026, where my doctoral research focused on developing and evaluating a low-cost, multi-camera markerless motion capture system for quantitative movement analysis. This work involved comparing markerless reconstruction methods against marker-based reference measurements and evaluating their performance for biomechanical outcomes, including gait kinematics, spatiotemporal parameters, and standing balance measures.

My research background combines biomedical instrumentation, quantitative data analysis, and scientific software development. During my master's research at Brown University, I developed and optimized an optical imaging system for neuroscience research, gaining experience in experimental measurement and biological signal analysis. My earlier research at Burke Neurological Institute also introduced me to applications of quantitative measurement in rehabilitation. More recently, I have contributed to the development of FreeMoCap and its supporting software, including tools for three-dimensional motion data processing and multi-camera recording.

I recently released an open validation dataset containing markerless and marker-based motion capture measurements collected during walking and standing balance tasks, together with analysis resources supporting reproducibility. This work has highlighted the importance of documenting measurement definitions, processing methods, and data provenance when interpreting and comparing biomechanical outcomes.

I am well suited to lead the proposed DAPR pilot project because it builds directly on these experiences. The project will harmonize existing motion capture datasets and extend FreeMoCap to support structured metadata capture, storage, and export. My experience with multiple pose estimation methods, biomechanical outcome measures, scientific software, and publicly shared research data provides a foundation for developing and evaluating these capabilities in collaboration with the DAPR Resource Core.

Honors

2016	Design Cultures and Creativity Capstone Award, University of Maryland
2016	Honors College Citation, University of Maryland
2014	National Merit Siemens Scholarship, National Merit Scholarship Corporation
2014 - 2018	Presidential Scholarship, University of Maryland

Contributions to Science

1. Development and maintenance of accessible, open-source motion capture

   Cost and technical complexity can limit access to quantitative movement analysis. I contribute to the development and maintenance of FreeMoCap, an open-source markerless motion capture system that uses synchronized consumer cameras to reconstruct three-dimensional human movement. My work spans the main FreeMoCap application and its supporting software libraries, including tools for camera recording, pose estimation, three-dimensional reconstruction, and biomechanical data processing.

   My contributions include developing features and analysis tools, resolving software defects and user-reported issues, managing releases and dependencies, and improving testing and application reliability. Together, these activities support FreeMoCap as a usable and adaptable research platform. Its open-source implementation allows researchers to inspect the methods used to generate movement measurements, modify processing workflows, and build on existing tools. My

development and maintenance work helps sustain these capabilities as the software evolves, supporting broader access to quantitative movement research and reproducible analysis.

2. Validation and reproducibility of markerless biomechanical measurements

   The usefulness of markerless motion capture depends on the accuracy of the measurements derived from it, which can vary across tracking methods, tasks, and outcomes. In my doctoral research, I evaluated FreeMoCap reconstructions using MediaPipe, RTMPose, and ViTPose against simultaneously recorded Qualisys marker-based measurements during treadmill walking and standing balance. The comparisons showed that performance depended on both the pose estimator and the measurement being evaluated. Methods that performed well for gait did not necessarily perform best for standing balance, demonstrating the importance of validating specific outcomes for their intended use.

   I developed the analysis workflows and evaluated agreement in joint kinematics, spatiotemporal gait measures, and center-of-mass-based balance measures. I also released the associated validation dataset and reproducible analysis resources, making matched markerless and marker-based trajectories available for independent reanalysis and method development. Together, these contributions provide evidence for interpreting FreeMoCap measurements and a reusable basis for evaluating changes to motion-capture and analysis methods.

3. Adapting markerless motion capture to prosthetic gait assessment

   General-purpose pose-estimation models can misidentify landmarks on prosthetic limbs, limiting their use for quantitative gait assessment. In a proof-of-concept study involving one participant with a transfemoral amputation, I developed and evaluated a hybrid approach combining RTMPose whole-body estimates with a DeepLabCut model trained to identify prosthetic landmarks. Comparisons with Qualisys marker-based measurements showed that the hybrid approach reduced prosthetic landmark error and recovered changes in selected gait measures across controlled prosthetic alignment adjustments. Performance remained dependent on the outcome, identifying limitations as well as improvements.

   My contributions included model development, integration of prosthetic landmarks into the reconstruction workflow, and biomechanical comparisons with the reference measurements. This work demonstrates how an open-source motion-capture workflow can be adapted to anatomical and device-related features that general-purpose models may not represent adequately. It provides a basis for further evaluation of markerless methods for prosthetic gait assessment. A manuscript describing this work is under revision.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

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Certified by Cherian, Aaron T in SciENcv on 2026-10-1613:06:21